

The Verified Constitutional Kernel

Machine-Checked Invariants of Axiomatic Humanist Cybernetics v3.1

Plain Language Summary · Formal Appendix · Reviewer Brief Kernel version 0.7 — 2026-07-12

Companion artifact to AHC v3.1 and Companion A (integrated). **89 theorems · 4 modules + a cross-module bridge · Lean 4.15.0, core only · zero unproven claims.** Source digest (sha256 over sorted sources):
8d262152c5f9036b20de9ea1c395a8edf31b4bfd2df893e8a4466d65b56736d0

Circulated under Principle 19.2 (Complexity Ceiling): this artifact adds no new specification. It verifies specification that already exists — including five rulings and two cross-module bridges arising from the first two review rounds — and identifies precisely where verification ends. Every change since the Phase 1 packet followed external review, never preceded it; the full change record is `ERRATA_AND_AMENDMENTS.md`, and every superseded version's digest is retained there for audit.

Part I — Plain Language Summary

This summary follows the framework's own Plain Language Output Layer requirement (AHC §10.4; Companion A Appendix B), applied to this artifact. A constitutional architecture that requires plain-language accountability of every output owes the same of its own proofs.

What this is, and what has been done

The AHC constitution contains rules written in careful English: rules about when the system may act and how hard, about how long an emergency may last, about how measurements are combined, about what the public record of a decision must contain. We translated four families of those rules into a language a computer can check with complete rigor — a proof assistant called Lean — and proved 89 statements about them. A proof assistant does not test a rule on examples; it verifies that no counterexample can exist, anywhere, ever, within the rules as written.

This is the seventh iteration of that kernel. The first (v0.1, 34 theorems) was circulated for review. Reviewers found no broken proof, but they found places where the English said more than the mathematics had captured, and questions the constitution had left open. Those findings were ruled on by the framework's stewards and the rulings were formalized — so the kernel now proves not only the original invariants but the resolutions of every question raised against it. The four families:

1. When the system may act, and how hard (the evidence ladder and the emergency clock). The system can never act more forcefully than its evidence permits, and everything it does on less-than-conclusive evidence can be undone. An emergency protective order that is not confirmed within 72 hours reverses itself automatically — and, as of this version, the 72-hour limit bounds the *evidence*, not merely the paperwork: a system cannot let a protective order lapse and immediately re-file the same stale case to buy another 72 hours. If real danger genuinely persists past the deadline, the system does not simply lose all protection (that would be inhumane) and does not silently keep its full emergency powers (that would be lawless). It enters a narrow **continuity-hold**: a floor of the mildest, fully reversible measures,

while a mandatory human review decides what happens next. Only genuinely new evidence — not the mere persistence of the alarm — can restart a full emergency order. *Continuity of a signal is never continuity of authority*. This is proved.

2. What "irreversible" means, and when the gravest tools are unlocked. The system's strongest enforcement mechanisms — cutting off capital, severing a community from part of the network — are not presumed harmless. Whether a given action is reversible is decided per deployment, by a public, contestable certificate that must account for how concentrated it is, how long it lasts, how critical the dependency, how fast it can be rolled back, and what cascades it could trigger. Reversibility is judged by whether the *affected people and institutions* can be restored — not by whether a wire can be reconnected. Without such a certificate, an action is treated as irreversible and requires the highest evidentiary tier (full causal proof). A deployment in which *no* amount of a given action is presumed safe — think payroll, medication supply, settlement infrastructure — is a legal configuration. The absence of a certificate is never evidence of reversibility.

3. How measurements survive corruption. If a minority of the system's sensors are captured or corrupted, they cannot drag its thresholds outside the range of honest measurements, and they can never fabricate a certified reading on their own. As of this version, the emergency system's alarm itself is derived from these same corruption-resistant measurements: a captured minority can neither manufacture the alarm that would keep a community under protective authority, nor stand as the sole basis of one.

4. What the public record of a decision must contain, and when. If any one community's viability collapses, the system's global health signal reads zero — no prosperity elsewhere can mask it (and we prove that a *sum* would allow masking where the chosen *product* does not; the design choice is what prevents it). The plain-language record and the technical record of any decision cannot disagree about anything that matters for challenging that decision while both comply with the rules — and if they do disagree, the disagreement itself proves which record broke the rules. Everything needed to challenge a decision is public at the moment the decision is made. As of this version, an emergency protective order's public record must additionally state, at hour zero: *why* it was authorized (new evidence, continuing danger, or both), *what makes any "new evidence" claim genuinely new* (and that determination can itself be challenged), *how many hours of unreviewed protection the incident has already consumed* — and that number provably matches the system's own internal count, so the public figure cannot be understated — and *what evidence would prove the order wrong*. These accountability fields, and the findings they accompany, must appear in *every* version of the public record — the immediate one and the later civic and technical ones alike; a later version cannot quietly drop what the first made public.

What this system of proofs can and cannot see

The proofs verify the rules as written — not the world the rules act on. They cannot see whether the sensors are truly independent, whether measurements track real harm, whether a reversibility certificate honestly describes reality, whether the recorded history of an incident is its true history, or whether a plain-language document genuinely means what its formal record claims. Those gaps are deliberate and named in the artifact as **the seam**: machines check everything that does not require human judgment, so that human judgment — exercised by affected communities through Layer 0 — is spent only where nothing can replace it.

What this artifact does not do right now

It does not make the AHC system exist, does not validate its statistical methods, and does not certify any deployment. It certifies that the constitutional text, in the parts formalized, cannot be lawyered into its opposite — and that the resolutions of the questions raised against earlier versions genuinely mean what they say.

What would prove this wrong, and how to challenge it

Every claim above is checkable by anyone, without trusting us. The complete source is included; one command (`lake build`) re-verifies every proof from scratch in seconds on an ordinary computer. If any proof were broken or missing, the build would fail and say so. The honest points of challenge are not the proofs but the modeling choices — the places where English was turned into mathematics — and the newer question of whether the certificate and continuity-hold designs place their trust in the right hands. Each module lists its own choices on its first page; Part III asks reviewers to attack exactly those.

If you find a problem — with a proof, with a translation choice, or with this document itself — Part III says exactly where it goes and who must answer it. A broken proof is fixed in the kernel; a wrong translation goes to the constitution's own amendment process; a problem with how burden is placed on communities goes to the Open Problems Register. Disagreement does not need our permission to be filed.

Where the full technical record is

The complete machine-checked source accompanies this document (`ahc-verified-kernel/`), with build instructions in its README and the theorem-to-specification map in Part II below. This summary is a navigation entry point, not a substitute for that record.

Part II — Formal Appendix

A. Scope and claim

The kernel formalizes the order-theoretic and counting-theoretic core of AHC v3.1: enforcement gating and the evidence–action ladder (§5.4, §9.2–9.3), emergency time and the crisis cap (Companion A §12.3, AHC §10.3, §4.2), measurement and aggregation (§8.3, §4.1–4.2, §1.1), and output-register invariance (§10.4; Companion A §5, §19.4, Appendix B). The claim is exact and limited: the specification, as modeled, provably possesses the 89 properties below. **The proofs verify the specification, not the world.**

Toolchain: Lean 4.15.0, core library only — no Mathlib, no external dependencies. **Zero sorry placeholders.** No theorem uses classical choice; the total axiom footprint is `propext` and `Quot.sound` (the standard Lean kernel axioms), and **40 of 89 theorems depend on no axioms at all**, including every Phase 1 theorem of the PLOL module. `decCompliant` is a verified `Decidable` instance, checked by elaboration and additional to the 81.

B. Module 1 — Tiered Evidence-Action Protocol (`TieredProtocol.lean`)

Phase 1 core (§5.4, §9.2–9.3).

Theorem	Verified guarantee
<code>tier_monotone</code> (T1)	Authorization is upward-closed: stronger evidence never removes an authorized response.
<code>severity_le_evidence</code> (T2)	Response severity never exceeds evidence rank: force cannot outrun proof.
<code>sub_causal_reversible</code> (T3)	Everything authorized below Tier 3 is reversible.
<code>irreversible_iff_causal</code> (T4)	Irreversible mechanisms are authorized exactly at full causal identification.
<code>broadcast_universal</code> (T5)	The transparency broadcast is authorized at every tier: the system can always say what it sees.
<code>pio_ceiling</code> / <code>pio_reversible</code> (T6, T7)	No PIO state authorizes above Tier-1 severity; everything a PIO authorizes is reversible.
<code>pio_resolves</code> (T8)	A PIO cannot remain pending past 72 hours: confirmed or auto-reversed (state-machine termination).
<code>no_chatter</code> (T9)	The hysteresis gap makes simultaneous escalate/de-escalate conditions unsatisfiable (instantaneous).

Temporal hysteresis — S1–S3 (§9.2; adopted v0.2, finding AHC-P1-002). T9 holds at an instant but is satisfied even by a zero-width band. S1–S3 state the *temporal* content of the gap: `oscillation_travel` (S1) — escalating and de-escalating values differ by more than the gap width; `flips_travel_anchored` (S2) — every posture flip costs the signal more than the gap of travel; `chatter_requires_travel` (S3, headline) — over any trajectory, $(\text{flips} - 1) \cdot (\text{gap} + 1) \leq \text{total variation}$, so sub-band noise can flip the enforcement posture at most once. Cycling requires genuine, repeated, full-band swings.

Episode machine, two clocks — E1–E11 (§5.4; adopted v0.4, rulings D-R1A + D-R4). The hazard clock (danger persists) and the epistemic clock (evidence advances) are separated. Only an attested materially new claim restarts a full 72-hour PIO; persistence of the alarm sustains only a constrained **continuity-hold** floor.

Theorem	Verified guarantee
<code>episode_no_relitigation</code> (E1)	Across any span with no novel evidence and no Layer 0 resolution, <i>whatever the exceedance and filing pattern</i> , unconfirmed full protection ≤ 72 hours.
<code>episode_single_issuance</code> (E2)	At most one full-PIO issuance per such span.
<code>expiry_routes_by_risk</code> (E3)	At the deadline: continuing risk $\rightarrow$ continuity-hold; subsided risk $\rightarrow$ spent. No third outcome, no discretion.
<code>reonsset_refloors</code> / <code>hold_persists</code> (E4, E5)	The hold floor follows the risk and is never withdrawn while risk persists unresolved.
<code>hold_resolution_iff</code> (E6)	Only a Layer 0 review output returns the hold to ordinary posture — not time, not filings.

Theorem	Verified guarantee
<code>novel_restart_from_spent</code> / <code>_from_hold</code> (E7)	An attested materially new claim restarts a full PIO.
<code>exceedance_cannot_restart</code> (E8)	Without novelty, no input re-enters the full clock: continuity of a signal is never continuity of the full authority.
<code>hold_floor_reversible</code> / <code>_severity</code> / <code>hold_grants_no_more_than_pio</code> (E9–E11)	The hold floor is reversible-only, Tier-1-capped, and a remnant of PIO authority — never an extension.

Certified reversibility envelopes — W1–W9 (§5.4, §9.3; adopted v0.4, rulings D-R2A + D-R3). An action carries an opaque deployment descriptor (magnitude, concentration, duration, rollback latency, cascade ...) judged by a deployment-certified `Envelope`. Routing *and* severance are irreversibility-bearing unless certified.

Theorem	Verified guarantee
<code>cert_tier_monotone</code> / <code>cert_severity_le_evidence</code> (W1, W2)	T1/T2 lifted to certificate-bearing actions.
<code>cert_sub_causal_reversible</code> (W3)	Everything authorized below Tier 3 is reversible — routing and severance — as a consequence of gating, not fiat.
<code>cert_irreversible_iff_causal</code> (W4)	T4 lifted to certificate-bearing actions.
<code>uncertified_routing_needs_causal</code> / <code>uncertified_severance_needs_causal</code> (W5a, W5b)	No certificate $\Rightarrow$ Tier 3. Q2 answered: uncertified severance is treated exactly as irreversible.
<code>certified_route_at_t1</code> / <code>certified_severance_at_t2</code> (W6a, W6b)	The graduated ladder is preserved where reversibility is demonstrated.
<code>cert_refinement_conservative</code> (W7)	The §9.3 mechanism table is a floor certificates can only raise, never lower.
<code>no_certificate_no_presumption</code> (W8)	Reversibility of routing/severance is exactly its certificate.
<code>zero_envelope_constructible</code> (W9)	A deployment with no presumptively reversible routing or severance is a legal parameterization.

C. Module 2 — Crisis Frequency Cap, Structural Review, Axiom II (`CrisisCap.lean`)

Phase 1 core + composition. `window_head_bound` (C1), `rolling_window_bound` (C2), `no_permanent_emergency` (C3): the rolling 24-month window carries at most $T_{\text{cap}} = 180$ emergency days, at every offset, for any continuation. `no_emergency_in_review` (C4), `review_absorbing` (C5), `review_gate` (C6): no event grants emergency authority during Structural Review, which only a Layer 0 constitutional output can close. `axiomII_dichotomy` (A2a): every episode is exactly one of reversibility-claim-valid or locally terminal.

Composed cap $\times$ review — G1–G7 (adopted v0.2, finding AHC-P1-001). The cap arithmetic and the review automaton, proved separately in Phase 1, are joined into one daily-step machine with the cap trip *derived* from the window arithmetic: `dayStep_valid` / `dayRun_valid` (G1, every reachable trace is protocol-valid, so C1–C3 apply verbatim), `review_day_never_emergency` (G2, an emergency day during review is unconstructible), `review_exit_iff_output` (G3), `trip_forced` (G4), `review_run` / `review_gate_composed` (G5),

`emergency_day_provenance` (G6, every emergency day was individually authorized),
`composed_cap_safety / composed_no_permanent_emergency` (G7).

D. Module 3 — Byzantine Measurement Consensus, Axiom I (`SensorsAndKernel.lean`)

Unchanged since v0.1. `honest_strict_majority` (B1), `exists_honest_le / _ge` (B2), `median_bracketed` (B3): under $m > 2f+1$, a captured minority cannot drag a median-type statistic outside the honest envelope. `no_corrupt_certificate` (B4): no coalition of $\leq f$ sensors can certify a reading no honest sensor made. `axiomI_null_kernel` (A1a) through `sum_admits_masking` (A1e): one non-viable subgraph nullifies global viability (intersection and product forms), the global signal is nonzero exactly when every subgraph's is, and aggregation by sum provably admits the Korematsu Illusion that the product rules out.

E. Module 4 — Plain Language Output Layer: Register Invariance (`PL0L.lean`)

Phase 1 core. `hash_bridge_origin` (P1), `tripartite_single_origin` (P1'), `bridge_proves_origin_not_consistency` (P2, the specification's own limitation proved with a witness), `compliant_registers_agree` (P3), `divergence_convicts` (P3'), `residual_divergence_harmless` (P4), `no_hostage` (P5), `contestability_at_breach` (P5'), `decCompliant` (P6): compliance is decidable; the divergence audit needs no expertise and no discretion.

D-R4 disclosures — P7–P9 (§10.4; adopted v0.5, ruling D-R4). A PIO-related event (`PIOEvent`) must carry the four D-R4 fields — authorization basis, novelty basis and lineage, cumulative episode protection, falsification condition — as mandatory members of the contestation-critical set; a withholding event is unconstructible.

Theorem	Verified guarantee
<code>pio_disclosure_at_breach</code> (P7)	All four D-R4 fields are in the T+0 semantic record at hour zero: contestation, including of the novelty determination itself, never waits.
<code>pio_disclosure_divergence_convicts</code> (P7')	Omission of any D-R4 field from any register proves that register non-compliant.
<code>attested_budget_accurate</code> (P8)	The disclosed budget figure carries a proof it equals the episode machine's accounting: given the incident history, a misstated figure is unconstructible.
<code>attested_budget_bounded</code> (P9)	Cross-module (Module 1 × Module 4): any attested budget over a novelty-free, review-free span is ≤ 72 hours, by E1. The public figure inherits the machine's constitutional bound.

Later-register invariance — P10–P13 (§19.4, App. B.4; adopted v0.7). Phase 1 obliged only the T+0 semantic record at release; the civic (T+72h) and technical (T+30d) registers are checked *when they ship*. `ShippedTripartite` carries their compliance as proof obligations.

Theorem	Verified guarantee
<code>tripartite_critical_consensus</code> (P10)	The three registers of a shipped compliant release agree on the <i>entire</i> contestation-critical fragment: one event, three voices, same findings.

Theorem	Verified guarantee
<code>later_registers_no_hostage</code> (P11)	Every critical claim is present in the civic and technical records too; a later register cannot drop what the semantic record carried.
<code>shipped_pio_disclosure_all_registers</code> (P12)	The four D-R4 fields appear in all three registers of a shipped PIO release, not the T+0 record alone.
<code>later_residual_divergence_harmless</code> (P13)	Any claim differing between the later registers is provably non-critical: differences are framing, never findings.

P9 is the kernel's first cross-module theorem: the disclosure layer and the authorization layer cannot drift apart on the one figure that measures how much unreviewed protection an incident has consumed.

F. Bridge — Exceedance Derivation (`ExceedanceBridge.lean` , Module 1 × Module 3)

The episode machine consumes `exceedance` — the hazard signal that sustains a continuity-hold — as an input. Adopted v0.6, this bridge derives it from the Byzantine measurement layer: a certified exceedance requires $f+1$ sensors reporting at or above the danger threshold, so by B2 (`exists_honest_ge`) it is always witnessed by an honest sensor.

Theorem	Verified guarantee
<code>derived_exceedance_honest_witnessed</code> (X1)	A certified exceedance always has an honest sensor at or above the danger threshold: the hazard signal is never a pure fabrication of the corrupted coalition.
<code>derived_exceedance_not_forgeable</code> (X2)	If every honest sensor reads below threshold, exceedance cannot be certified: a captured minority cannot raise the alarm alone.
<code>hold_sustained_only_by_witnessed_danger</code> (X3)	Whenever the machine keeps a subgraph in a continuity-hold on a derived input, honest sensors corroborate the danger.
<code>manufactured_danger_cannot_sustain_hold</code> (X4)	Fabricated alarm drops the subgraph out of the hold to <code>spent</code> : the laundering path closed at the measurement layer.

Together with E1/E8 (the authorization layer) and the D-R4 disclosures (the record layer), this closes the emergency loop's last free input: alarm, authority, and disclosure are now grounded in one another, and no one of them can be manufactured without the others.

G. Modeling disclosures — the contestation targets

Per the framework's norm of separating the contestable normative layer from the formal one, every translation choice is stated in the header of its module. These, not the proofs, are the right places to aim review. The most load-bearing:

- **Time is discretized** — hours for the PIO/episode machine, days for the Crisis Cap trace; the 72h, 180-day and 24-month constants are the spec's.
- **The episode's `novel` input is an ATG/Layer 0 attestation** attached to a claim identity (D-R4): rewording or repetition is not novelty, and the determination is contestable through Layer 0, made *outside* the machine. `exceedance` is the raw risk signal. The two are separate inputs by design — the hazard clock and the epistemic clock never share a needle.

- **The continuity-hold floor is a deployment-certified `HoldPolicy`**, constrained by construction to reversible mechanisms at or below Tier 1. Which measures constitute the floor is a deployment/Layer 0 determination.
- **Reversibility is an opaque deployment certificate (`Envelope`)** over a descriptor the kernel does not interpret; the kernel gates on the certificate but does not verify that it describes reality. The v0.3 scalar bound is the special case $\delta := \mathbb{N}$, `routeInside := ( · ≤ bound )`; no positivity survives.
- **A budget attestation carries its own proof** of equality with the episode machine's count; whether the *recorded history* is the true history is institutional (append-only logs, Layer 0 audit).
- **Constitutive constraints are structure fields** — the hysteresis gap, $T_{\text{cap}} < W$, the $m > 2f+1$ quorum, the mandatory D-R4 disclosure fields — so degenerate or withholding parameterizations are unconstructible, not merely invalid.

H. What is not verified, and why

The statistical layer (three-stage causal identification §5.5, synthetic control validity, capture-rate estimation) is not Lean-shaped; its instrument is pre-registration and blind historical validation (OP.4, Companion B). Viability-kernel dynamics over continuous state spaces are the province of viability theory. Sensor independence is an institutional achievement (OP.1), assumed here as the $\leq f$ fault bound. The certificate and continuity-hold designs move trust to new institutional points — who issues, contests, and revokes an `Envelope` or `HoldPolicy`, and who guarantees an incident's recorded history — and those points are named, not resolved, here. And the relation between a natural-language text and the claim set it expresses — the seam — is formally unverifiable by construction: all trust is consumed at claim extraction, none after it, and extraction belongs to Layer 0. The kernel's contribution to that division of labor is to prove that everything after extraction is mechanical (`decCompliant`), that divergence convicts (`divergence_convicts`, `pio_disclosure_divergence_convicts`), and now that the one quantitative disclosure cannot lie (P8, P9).

I. Reproduction

Requirements: any machine with `elan` (the Lean toolchain manager). Then:

```
cd ahc-verified-kernel
lake build
```

The build elaborates every proof and prints the axiom audit (`AHCKernel/Audit.lean`) to the log. A missing or broken proof fails the build; a placeholder surfaces as the `sorryAx` axiom. **Expected:** 89 audited theorems, each at most `[propext, Quot.sound]`; never `Classical.choice`; 40 axiom-free. A continuous-integration workflow re-runs this check on every change and fails on any `sorry`, any classical choice, any unexpected axiom, or any deviation from the 89/40 footprint.

Part III — Reviewer Brief

Companion A Principle 19.2 holds that further specification must follow, not precede, external review. This artifact complies: the first two review rounds produced findings (AHC-P1-001..008; CG-1..CG-6), the framework's stewards ruled on them (D-R1A, D-R2A, D-R3, D-R4), and the rulings were formalized. This round solicits attack on the results of that process, and on the surfaces the newest work exposes. Findings at any of the following addresses are the intended product of circulation.

For formal-methods reviewers

F-1 (Faithfulness of the rulings). For each ruling now in the kernel, does the formal reading weaken or strengthen the constitution's intent? The strongest finding is a behavior satisfying the Lean model while violating the English, or vice versa. Specific targets: does the episode machine's routing at expiry (E3) match §5.4's intent for an unconfirmed order under persisting danger? Does `exceedance_cannot_restart` (E8) forbid any legitimate re-escalation the spec intends?

F-2 (The continuity-hold semantics). The hold is modeled as a single per-subgraph mode with a `HoldPolicy` floor. Does this admit any gaming path — e.g., alternating hold/spent to reset an implicit clock, or a `HoldPolicy` that is nominally reversible yet materially punitive? Is there a bound the hold *should* carry (a maximum hold duration before mandatory resolution) that it does not?

F-3 (Certificate expressiveness). Envelopes are opaque predicates over an abstract descriptor. Is that the right altitude, or should the kernel constrain the descriptor's structure (e.g., require monotonicity in a stated coordinate, or compositionality under combined actions)? Does anything in the spec's intent survive only if the certificate carries more structure than a bare Boolean?

F-4 (Strengthening candidates). The two candidates named in the previous brief are now discharged: `exceedance` is derived from Module 3 (v0.6, X1–X4, Part II.F) and later-register compliance is proved (v0.7, P10–P13). The order-theoretic surface that remains is thinner, and the highest-leverage open questions have shifted from the formal layer to the institutional one (D-1..D-4 below). A reviewer who still sees an order-theoretic sentence of §5.4, §12.3, §19.4, or Appendix B that could join the kernel — and survive Principle 19.2 — should name it here; that is now the scarcer finding.

For constitutional and domain reviewers

D-1 (Certificate governance). Rulings D-R2A/D-R3 place reversibility in a deployment certificate that must be public, contestable, and revocable. Who issues it, who may contest it, and what happens to actions taken under a certificate later revoked? This is the largest open institutional surface the kernel opens and does not close.

D-2 (The novelty determination). D-R4 makes `novel` a contestable Layer 0 attestation. In practice, who adjudicates novelty under time pressure during an ongoing incident, and does the contestability right function before, not only after, the protective window it governs?

D-3 (Hold humaneness in deployment). The continuity-hold prevents an abrupt protection vacuum in theory (E5). Does a real deployment's `HoldPolicy` floor actually protect an affected community, or could a minimal floor satisfy

the theorem while leaving people materially unprotected — reproducing the lock-without-key pattern one level down?

D-4 (Severance certification in the field). D-R3's soft-severance conditions (time-bounded, state-preserving, essential-service floor, tested restoration, cascade-safe, repair mechanism) are constitutional text. Are they the right conditions, and is "tested restoration rather than theoretical reconnection" operationalizable for the deployments AHC intends?

For community and legibility reviewers

L-1 (Reflexive compliance). Part I claims PLOL conformance for this document. Audit it against Appendix B.2 and the new D-R4 fields: is anything a community reviewer needs missing, and does the falsification pathway function for a non-expert with the Community Statistical Support Fund's resources?

L-2 (The budget figure's meaning). P8/P9 prove the disclosed episode budget cannot understate the machine's count. But the figure is only as honest as the recorded incident history. Is "the recorded history is the true history" a burden that quietly falls on communities, and what institutional guarantee (append-only logs, independent Layer 0 audit) does it require?

L-3 (The seam's placement, revisited). Every ruling this cycle moved trust to a new named point — certificate issuance, novelty attestation, history integrity. Are these the right custodians, or does the accumulation relocate expert burden onto communities across several fronts at once?

Disposition of findings

Findings against proofs (F-class) invalidate specific theorems and are corrected in the kernel. Findings against modeling choices and rulings (F-1, D-class) are constitutional-interpretation questions and route to the specification's amendment processes; the kernel then re-verifies the amended reading — as it did five times to reach this version. Findings against the seam's placement (L-class) route to the Open Problems Register. In all three cases the artifact's function is the same: it converts disagreement about what the constitution means into a precise, checkable object — which is what a formal layer is for.

— End of circulation brief. The machine-checked source, the module READMEs, the theorem-to-specification map, the amendment record (`ERRATA_AND_AMENDMENTS.md`), the archived external reviews (`external-reviews/`), and the CI workflow accompany this document. —